

Protocol for the Experiment
X-ray Fluorescence
(Experiment 256)

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Preface and Notes

Use of External Tools

For the linguistic drafting and correction of the Introduction (Kapitel ??) and the Execution (Kapitel ??), the AI *Lumo AI* as well as *Duden Mentor* were employed. However, the substantive evaluation and the actual creation of the document were carried out entirely without the assistance of Artificial Intelligence, unless explicitly marked otherwise. [Dud26, Pro26]

All graphics contained in the document that lack a specific reference were independently created or edited in Inkscape, or are plots generated in Python. Both the Python code and the LaTeX document were prepared in the Visual Studio Code (VS Code) development environment.

Own Code and Transparency

To carry out the evaluations, a custom Python library was developed. The entire source code - both the Python scripts and the LaTeX sources - is self-created and publicly available online.

The entire project structure can be found on my GitHub. Here, both the Python code and the LaTeX document code are documented. [Fin26a]

The Python functions used are specifically identifiable on GitHub under the name "Papulator" [Fin26b]

Use of University Materials

This document is based on the script by Jens Wagner (as of April 27, 2026). Graphics from this script were adopted for the Introduction. These excerpts are transparently marked; the figures used always originate from the chapters of the script that share the same experiment number and experiment name as in the present document. In this experiment, the graphics from *Experiment 256 X-ray Fluorescence* were used. [Wag26]

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1. Introduction

1.1 Task and Motivation

Radioactive decay represents one of the fundamental phenomena in nuclear physics, demonstrating the stochastic nature of quantum mechanical processes. The primary objective of this experiment is to investigate the statistical properties of radioactive decay by measuring count rates over time and analyzing their distribution characteristics. Understanding the statistical behavior of radioactive emissions is essential not only for theoretical physics but also for practical applications in radiation protection, medical diagnostics, and nuclear instrumentation. The motivation behind this investigation stems from the need to verify the Poissonian nature of decay events and to develop proficiency in handling measurement uncertainties inherent in counting experiments. Additionally, this experiment provides insight into background radiation contributions and methods for their proper subtraction when determining the activity of radioactive samples. By systematically varying measurement parameters such as counting time and distance from the source, we aim to establish the relationships governing radioactive decay statistics and validate the theoretical predictions derived from first principles.

1.2 Physical Basics

1.2.1 Moseley's Law for Fluorescence

X-ray fluorescence occurs when incident photons, particularly X-rays, excite atoms in a material, causing electrons to be promoted to higher energy levels. When these excited atoms return to their ground state, they emit photons whose energy corresponds to the difference between the relevant energy levels. To predict the energy of the resulting photons, an expression for the energy differences between atomic states is required. The Bohr atomic model provides a framework for approximating the energy E between states with principal quantum numbers n and m :

$$E = E_R \left(\frac{(Z - \sigma_n)^2}{n^2} - \frac{(Z - \sigma_m)^2}{m^2} \right), \quad (1.1)$$

where Z represents the atomic number and E_R denotes the Rydberg energy with a literature value of approximately $E_R \approx 13,6$ eV. The screening constants σ_i account for the shielding of the nuclear charge by surrounding electrons. For practical purposes in this experiment, the screening constants are combined into a mean value σ_{nm} , yielding the simplified expression

$$E = E_R (Z - \sigma_{nm})^2 \left(\frac{1}{n^2} - \frac{1}{m^2} \right). \quad (1.2)$$

Rearranging this equation leads to Moseley's law in its standard form:

$$\sqrt{\frac{E}{E_R}} = (Z - \sigma_{nm}) \sqrt{\frac{1}{n^2} - \frac{1}{m^2}}. \quad (1.3)$$

For this experiment, transitions from $n = 2$ and $n = 3$ to the ground state ($m = 1$) are of primary interest, corresponding to the K_α and K_β spectral lines respectively. For K_α radiation (transition $2 \rightarrow 1$) and not too heavy nuclei with $Z \approx 30$, the screening constant can be estimated as $\sigma_{12} \approx 1$, giving

$$\sqrt{\frac{E}{E_R}} = (Z - 1) \sqrt{\frac{3}{4}}. \quad (1.4)$$

By plotting $\sqrt{E}$ as a function of the atomic number Z and fitting Moseley's law to the measured data, both the Rydberg energy E_R and the screening constant σ_{nm} can be determined experimentally and compared to established literature values. This verification demonstrates the validity of the hydrogen-like approximation for inner-shell electron transitions in multi-electron atoms.

1.2.2 Semiconductor Detector

To detect the photons produced by X-ray fluorescence, semiconductor detectors are employed. These devices operate based on a pn-junction where an external voltage is applied. When n-type and p-type semiconductors are brought into contact, free charge carriers at the junction annihilate, creating a region devoid of free charges known as the depletion region. This depletion region can be expanded by applying a reverse bias voltage across the pn-junction, which pulls free charge carriers toward the exterior and widens the zone where charge carriers are scarce. The depletion region constitutes the sensitive volume of the detector. When an X-ray photon traverses this region, it can be absorbed through the photoelectric effect, ionizing atoms within the depletion zone. Due to the applied voltage across the pn-junction, the created charge carriers are accelerated, producing a measurable current pulse at the edges of the depletion region. Crucially, the number of free charge carriers produced is proportional to the energy of the incident photon, meaning the amplitude of the resulting pulse allows reconstruction of the photon energy. The

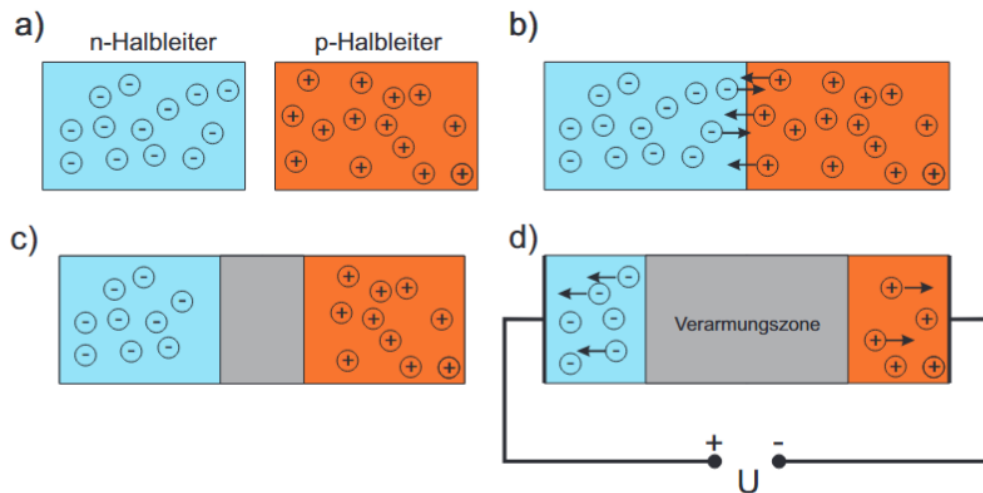


Figure 1.I:

Schematic depiction of a semiconductor detector showing the pn-junction, depletion region, and charge carrier movement upon X-ray photon absorption

relationship between deposited energy and signal amplitude requires a highly sensitive detector system for accurate measurement. An integrator circuit is typically used to accumulate the charge generated by each detection event, producing a voltage pulse proportional to the incident photon energy. This linear relationship enables the construction of energy spectra through multichannel analysis, where detected events are sorted into channels according to their pulse heights. The resolution of such detectors depends on factors including the semiconductor bandgap, temperature, and electronic noise characteristics, all of which must be controlled to achieve precise energy measurements suitable for elemental identification via characteristic X-ray line analysis.

Experiment V256 - X-Ray fluorescence

Date: 30.06.2026

Annika Künstle, Finn Ziemer, Scott Riley

Table I) Centres of K_{α} -lines in "Channel-Units"

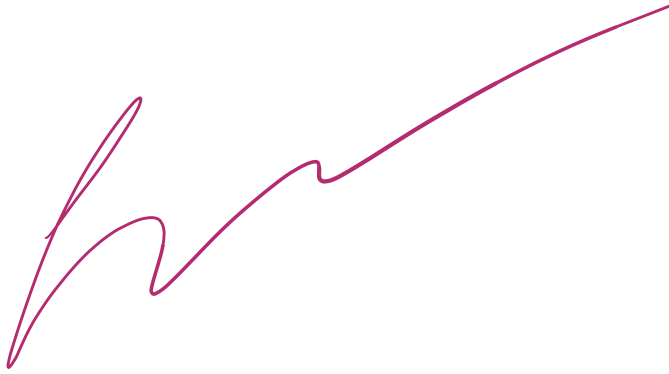
Element	Centre μ	σ
Fe	85	2
Mo	230	2

Table II) Every element with the measured K_{α} - and K_{β} -lines

Element	K_{α} [keV]	K_{β} [keV]
Fe #1	6.40 ± 0.16	7.03 ± 0.18
Mo #2	17.45 ± 0.18	19.54 ± 0.23
Ag #6	2.80 ± 0.13	24.53 ± 0.14
Cu #7	8.05 ± 0.17	8.89 ± 0.20
Ni #8	7.47 ± 0.17	8.26 ± 0.18
Ti #9	4.67 ± 0.16	Could not be resolved
Zn #10	8.65 ± 0.17	9.58 ± 0.16
Zr #11	15.87 ± 0.17	17.68 ± 0.20

Table II) Elements contained in the alloys

Alloy		Elements
1	#12	Shown
2	#13	in
3	#14	Graphs
4 (avg)	#15	



3. Procedure

3.1 Measurement Method

The experimental investigation employs a Geiger-Müller counter system configured for counting ionizing radiation events from a radioactive source. Each decay event that triggers the detector produces a distinct electrical pulse, which is registered by the connected counting electronics. The fundamental measurement principle involves recording the number of pulses accumulated during predefined time intervals, thereby converting the inherently continuous decay process into discrete count data suitable for statistical analysis. Multiple series of measurements are conducted under varying conditions to characterize different aspects of the decay statistics. The primary measurement series consists of repeated consecutive countings with fixed duration to establish the count distribution and verify its conformity to Poisson statistics. Additional measurement sequences explore the temporal evolution of count rates over extended periods to assess stability and drift effects. To quantify background contributions, dedicated background runs are performed without the radioactive source present, capturing ambient radiation levels from cosmic rays and environmental radioactivity. Geometric dependencies are investigated by systematically varying the source-to-detector distance while maintaining all other parameters constant, allowing verification of the inverse-square law behavior. Throughout all measurements, careful attention is paid to avoiding parallax errors, ensuring reproducible positioning, and minimizing external disturbances that could compromise measurement integrity. The collected raw data consist of count numbers paired with their corresponding accumulation times, forming the basis for subsequent uncertainty analysis and statistical testing.

3.2 Experimental Setup

The measurement apparatus comprises several interconnected components assembled to facilitate accurate detection and registration of ionizing radiation. The central detection element is a Geiger-Müller tube housed within a protective shielding enclosure, designed to maximize sensitivity to beta and gamma radiation while minimizing interference from charged particles. Connected to the detector head is a high-voltage power supply that establishes the necessary electric field within the gas-filled tube for avalanche multiplication of ionization electrons. The pulse output from the detector feeds into a scaler-timer unit capable of accumulating counted events and displaying elapsed measurement durations with digital precision. All electronic components are interconnected using standardized coaxial cables to minimize signal loss and electromagnetic interference. The radioactive source, securely mounted on an adjustable stand, can be positioned at precisely controlled distances from the detector window through a calibrated translation stage. Lead shielding blocks are available for background measurements and for attenuating stray radiation during alignment procedures. The entire assembly rests on a vibration-damped optical table to prevent mechanical perturbations that might affect geometric reproducibility. Power connections are organized to allow emergency shutdown capability while maintaining stable operating voltages throughout extended measurement sessions.

3.3 Spectrum for each single element

For this protocol eight different elements have been evaluated. All eight elements and their measured spectrum can be found in figure 3.II

The measured elements are (Fr, Mo, Ag, Cu, Ni, Ti, Zn, Zr) – with their atomic number $Z = (26, 42, 47, 29, 28, 22, 30, 40)$.

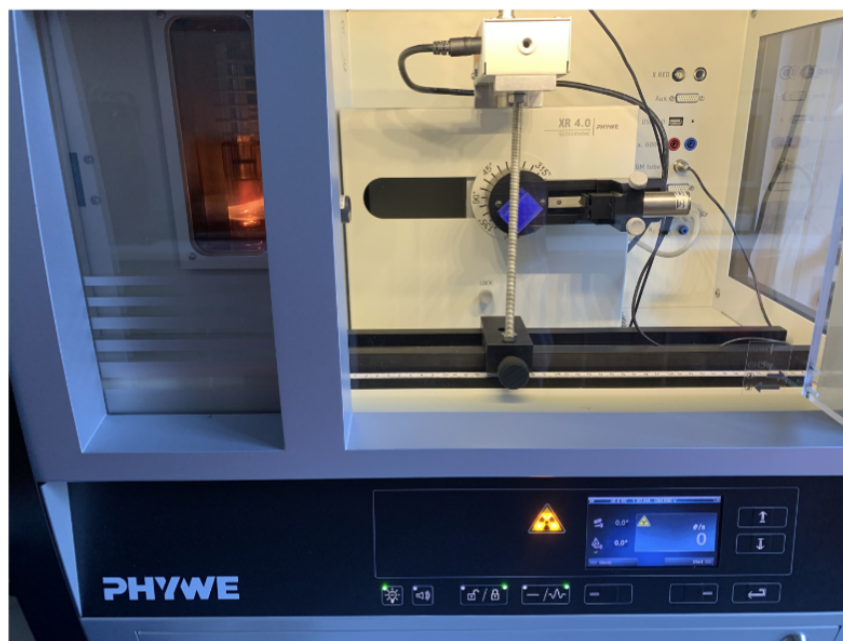


Figure 3.I:
Experimental setup

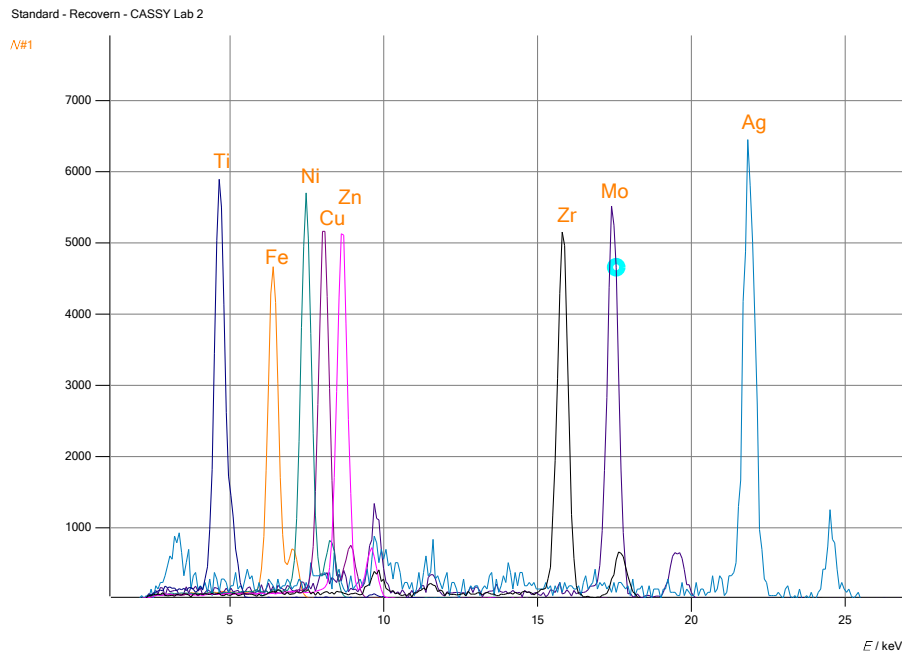


Figure 3.II:
All elements and their spectrums.

4. Evaluation

Uncertainty Calculation

For the statistical evaluation of n measurements x_i , the following equations are defined: [Wag25]:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{Arithmetic Mean} \quad (4.1)$$

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{Variance} \quad (4.2)$$

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{Standard Deviation} \quad (4.3)$$

$$\Delta \bar{x} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n (\bar{x} - x_i)^2} \quad \text{Uncertainty of the Mean} \quad (4.4)$$

$$\Delta f = \sqrt{\left(\frac{\partial f}{\partial x} \Delta x\right)^2 + \left(\frac{\partial f}{\partial y} \Delta y\right)^2} \quad \text{Gaussian Error Propagation for } f(x, y) \quad (4.5)$$

$$\Delta f = \sqrt{(\Delta x)^2 + (\Delta y)^2} \quad \text{Uncertainty for } f = x + y \quad (4.6)$$

$$\Delta f = |a| \Delta x \quad \text{Uncertainty for } f = ax \quad (4.7)$$

$$\frac{\Delta f}{|f|} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \quad \text{Relative Uncertainty for } f = xy \text{ or } f = x/y \quad (4.8)$$

$$\sigma = \frac{|a_{lit} - a_{gem}|}{\sqrt{\Delta a_{lit}^2 + \Delta a_{gem}^2}} \quad \text{Significant Deviation Calculation} \quad (4.9)$$

4.1 Evaluation of the alpha lines

In this section the aim is to verify Mosely's Law by evaluating E_R by using the K_α lines from different elements. Each element has a unique atomic number Z , given by the amount of protons in the nucleus. The root of the measured energy is plotted against the atomic number Z for each element. This can be found in figure 4.I. Using equation 1.2, a regression can be fitted to the measurement data. The regression and its parameters can be found in the same plot (4.I) respectively.

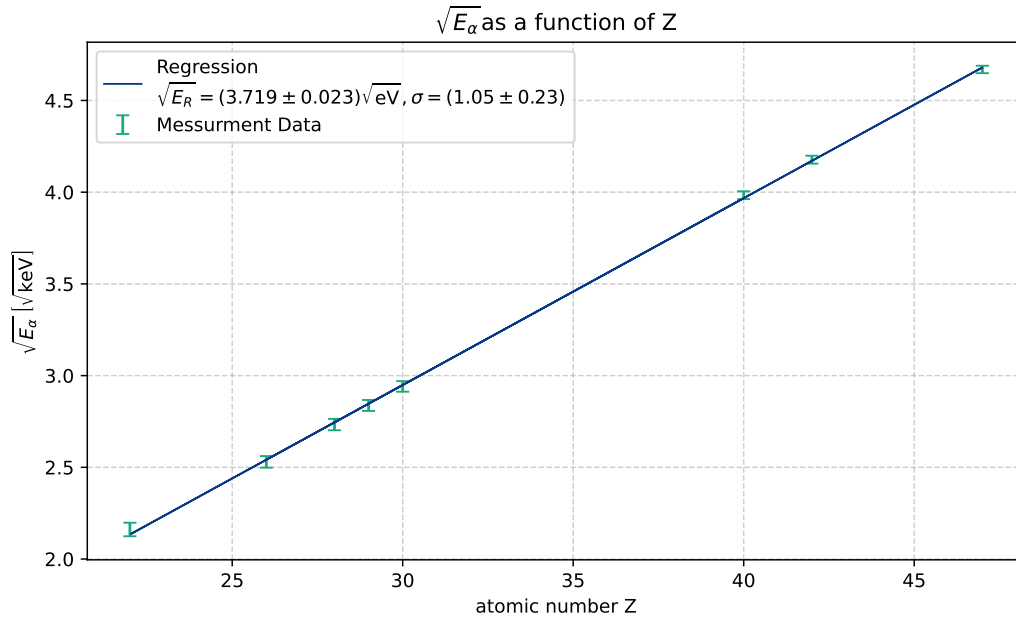


Figure 4.I:

Square roots of the measured energies against the atomic number, also containing the (linear) Regression for the K_α -lines.

For the regression n and m were set to $n = 1$ and $m = 2$. To get now the energy E_R for Moseley's Law the fit parameter $\sqrt{E_R} = (3.719 \pm 0.023)\sqrt{\text{eV}}$ will be squared:

$$E_R = \sqrt{E_R}^2, \quad \Delta E_R = 2 \left(\Delta \sqrt{E_R} \right) \sqrt{E_R}. \quad (4.10)$$

the uncertainty was estimated by using the [Gaussian error propagation \(4.5\)](#). Having all the data the energy can hence be solved to be

$$E_R = (13.83 \pm 0.17) \text{ eV} \quad (4.11)$$

The literature value for the energy of Moseley's Law is $E_R = 13.606 \text{ eV}$, hence the [standard deviation \(4.3\)](#) is 1.3σ and therefore statistically not significant.

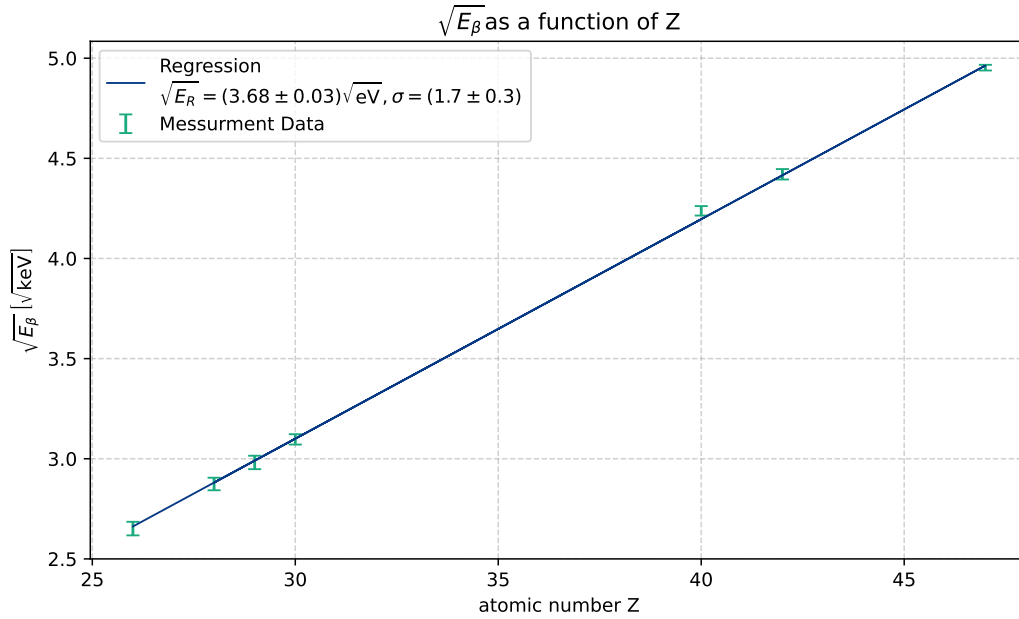
The second fit parameter is

$$\sigma_{12} = (1.05 \pm 0.23) \quad (4.12)$$

and often is approximated to be equal to one. The [standard deviation \(4.3\)](#) from the estimated value to one is just 0.21σ and hence statistically not significant, thus the approximation is reasonable.

4.2 Evaluation of the beta lines

The same analysis as in the section before will be done here, but instead analyzing the K_β -lines. The measurement data is plotted in figure 4.II.

**Figure 4.II:**

Square roots of the mussured energies against the atomic number, also containing the (linear) Regression for the K_β -lines. Ti was removed, since the value for this element could not be measured.

For the regression n and m were set to $n = 1$ and $m = 3$. Since the energy line for Ti could not be resolved, this values was removed from the analysis. The regression parameters as well as the plotted regression can again be found in the same figure respectively. Using equaiton 4.10 the enegy E_R can be calculated to be

$$E_R = (13.51 \pm 0.24) \text{ eV} \quad (4.13)$$

This value has a [standard deviation \(4.3\)](#) from the literature value of 0.39σ .

The other fit parameter has been evaluated to be

$$\sigma_{13} = (1.7 \pm 0.3), \quad (4.14)$$

having a standard deviation to one of 2.07σ . Beign still statistically not significant the approximation is not as good anymore and should be avoided, or only be used for rough approximations.

4.3 Evaluation of elements contained in each alloy

Compositional analysis of the unknown alloys documented in the measurement log is evaluated in this section. Measured fluorescence spectra are presented in the accompanying figures alongside the expected characteristic lines of elements hypothesized to constitute each alloy. Identification was performed by manually overlaying the expected spectral lines of various candidate elements and selecting the combination that provided the best match to the measured spectrum.

This empirical approach carries notable limitations. Time constraints of the experiment precluded systematic comparison of all plausible elemental combinations, rendering the identification process inherently subjective. Consequently, the assignment of specific spectral features to particular elements relies on visual judgment rather than automated fitting procedures.

However, some alloys could be figured out quite precisley, namly, alloys one and two. The third alloy has a lot of noise on the lower energy side, that could not be explained. For alloy four is was the same for high energy, as wlel as the most right peak, there was no (non-toxic) element, that fitted this peak.

Also in alloy 5, plenty of elements fit the peaks, but are considered toxic and thus unlikely to be used in an university experiment (hopefully).

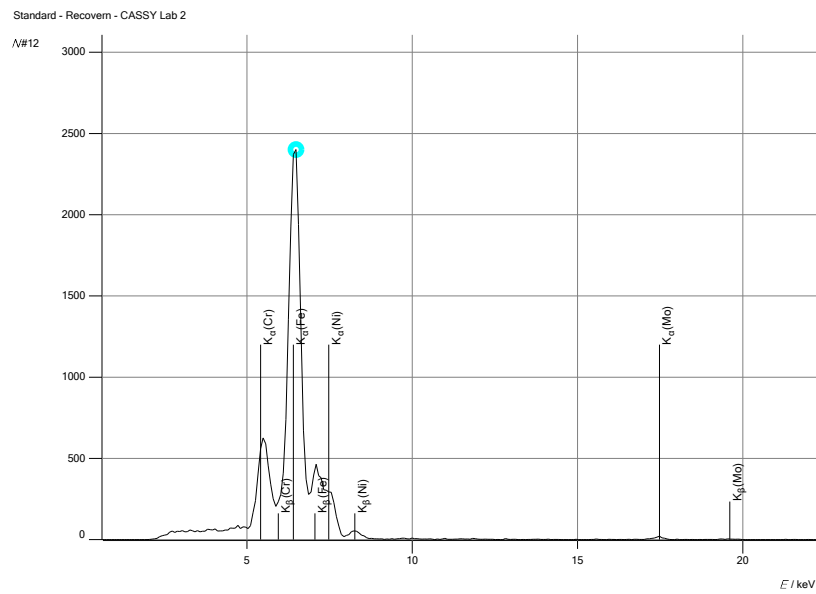


Figure 4.III:
Messured spectrum for alloy 1.

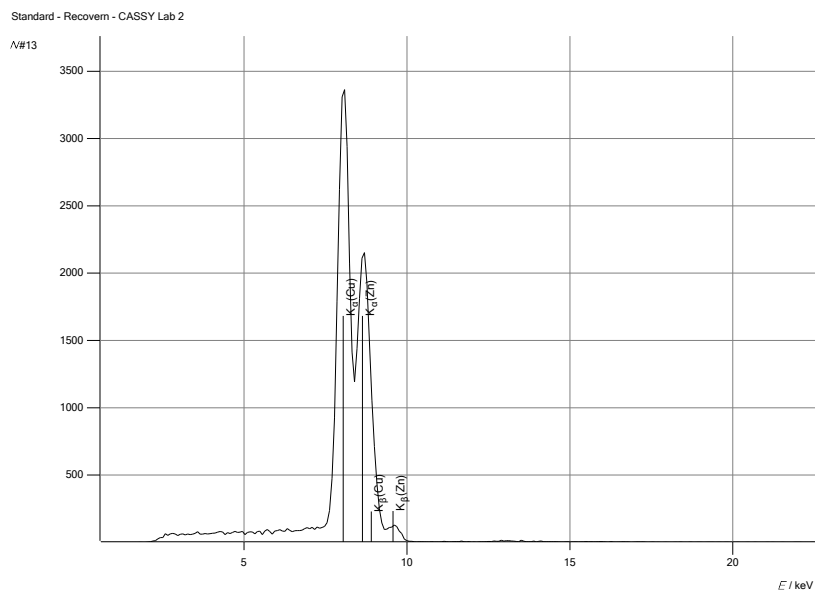


Figure 4.IV:
Measured spectrum for alloy 2.

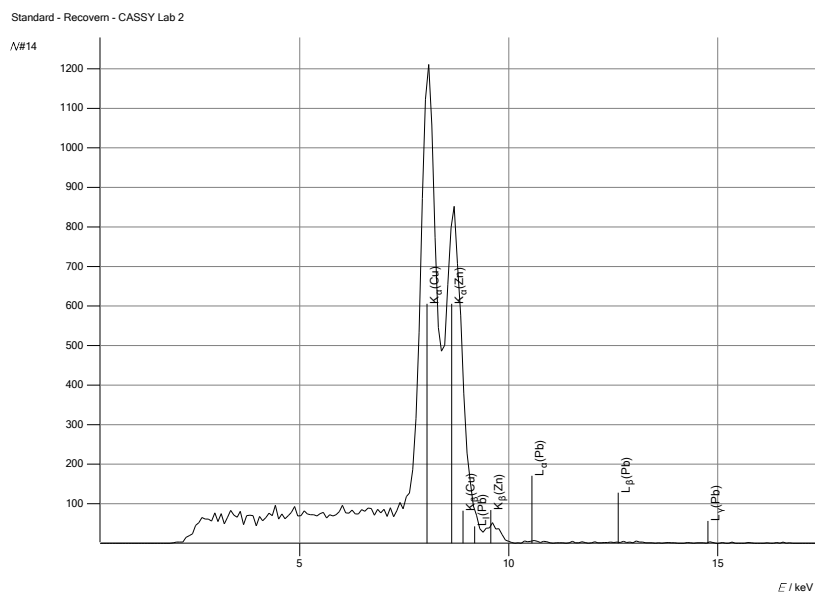


Figure 4.V:
Measured spectrum for alloy 3.

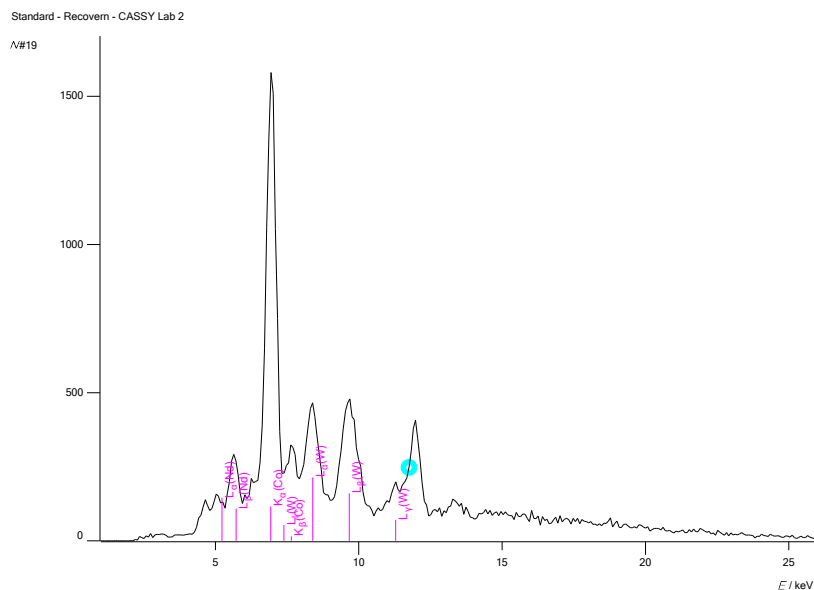


Figure 4.VI:
Measured spectrum for alloy 4. This was the magnet.

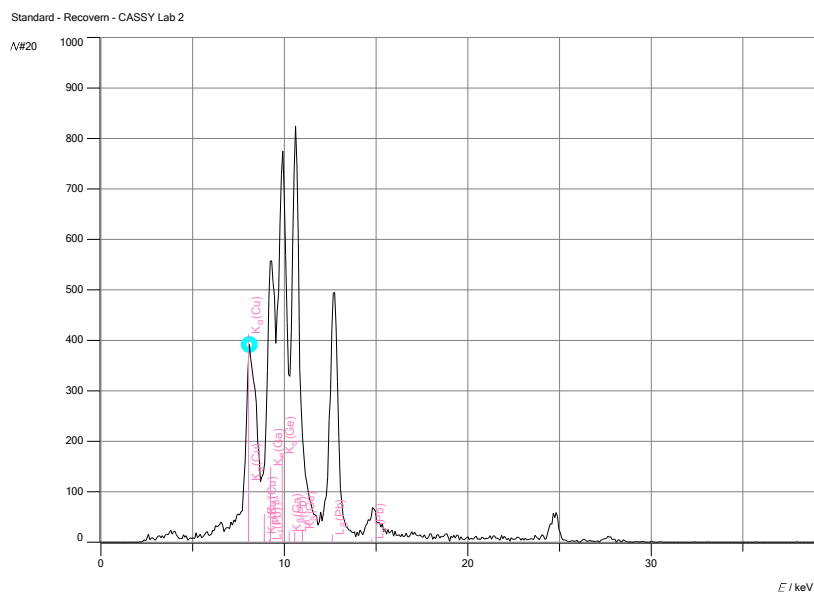


Figure 4.VII:
Measured spectrum for alloy 5.

5. Discussion

5.1 Summary

The aim of this experiment was to investigate the characteristic X-ray fluorescence spectra of different elements and to verify Moseley's law. After calibrating the detector using the characteristic emission lines of iron and molybdenum, the characteristic K_α and K_β lines of several elements were measured. From these measurements, the Rydberg energy and the corresponding screening constants were determined by linear regression according to Moseley's law. In addition, the fluorescence spectra of several unknown alloys were analyzed in order to identify their elemental composition by comparison with expected characteristic emission lines.

Overall, the measured data are in good agreement with the theoretical predictions. The fitted Rydberg energies obtained from both the K_α and K_β series agree with the literature value within their experimental uncertainties. Furthermore, the composition analysis of the unknown alloys produced plausible results, although the identification procedure was limited by the manual evaluation of the spectra.

5.2 Analysis of the Data

The evaluation of the K_α lines yielded a Rydberg energy of

$$E_R = (13.83 \pm 0.17) \text{ eV},$$

which differs from the literature value by only 1.3σ . Since this deviation is statistically insignificant, the experimental data confirm Moseley's law within the measurement uncertainty. Likewise, the fitted screening constant

$$\sigma_{12} = (1.05 \pm 0.23)$$

is fully consistent with the commonly used approximation $\sigma_{12} \approx 1$. This demonstrates that the simplified form of Moseley's law provides an accurate description of the measured K_α transitions.

For the K_β lines, the obtained Rydberg energy

$$E_R = (13.51 \pm 0.24) \text{ eV}$$

also agrees with the literature value, exhibiting a deviation of only 0.39σ . Although the determination of the Rydberg energy remains reliable, the fitted screening constant

$$\sigma_{13} = (1.7 \pm 0.3)$$

shows a considerably larger deviation from the approximation $\sigma_{13} = 1$. Its relative deviation exceeds 70%, indicating that this approximation is not suitable for quantitative analyses of the K_β transitions.

Several factors may explain this behaviour. First, the K_β emission lines possess significantly lower intensities than the corresponding K_α lines, making them more susceptible to statistical fluctuations, detector noise, and background radiation. Secondly, Moseley's law assumes a constant screening parameter for a given transition. In reality, the screening depends on the electronic structure of the atom and therefore cannot generally be treated as a universal constant. Consequently, deviations are expected to become more pronounced for the K_β transitions.

The analysis of the unknown alloys was performed by comparing the measured fluorescence spectra with the expected characteristic emission lines of possible constituent elements. This approach yielded convincing results for alloys with relatively simple compositions, where the characteristic peaks could be assigned unambiguously. More complex spectra, however, contained overlapping peaks or unexplained features that prevented a unique identification of all elements. Nevertheless, the proposed compositions are physically reasonable and consistent with the measured spectra.

5.3 Criticism

Several sources of uncertainty influenced the accuracy of the experiment. The energy calibration directly affects all subsequent measurements, so even small calibration errors propagate into the determination of the Rydberg energy and the screening constants. In addition, the finite energy resolution of the detector limited the separation of closely spaced spectral lines, especially for weaker K_β transitions.

The identification of the unknown alloys represents another important limitation. Since the elemental composition was determined by manually comparing measured spectra with expected characteristic lines, the analysis is inherently subjective. A more rigorous approach would involve automated peak fitting together with a quantitative least-squares analysis or comparison with reference spectra. Such methods would significantly reduce the influence of personal interpretation and improve the reproducibility of the results.

Overall, despite these experimental limitations, the obtained results are in good agreement with the theoretical expectations. The experiment successfully verifies Moseley's law within the experimental uncertainty and demonstrates the suitability of X-ray fluorescence spectroscopy for qualitative elemental analysis.

6. Python

The entire Python code can be found on my GitHub at <https://github.com/FinnZeumer/PAP-2> [Fin26a]. Additionally, the source code for this project itself is also available there, should anyone be interested in viewing it.

The code is primarily self-written. However, the built-in "MS Copilot" function in Visual Studio Code is used in parts. Unless otherwise indicated, it is used as a support tool and not to generate entire methods. Furthermore, we have a shared folder within our friend group where we share all of our code projects.

PAP 2.2 – Python-Auswertung

Wichtige Funktionen für das PAP.

Alle in diesem Dokument geschriebenen Zeilen Code sind von mir eigenständig geschrieben. Für ein sauberes und strukturiertes Dokument wurde eine eigene Pythonlibrary geschrieben, welche insbesondere ans PAP angepasst wurde. Dieses ist unter dem Namen "Papulator" auf meinem GitHub zu finden. Diese soll stetig ausgearbeitet werden. Der Stand dieses Protokolls ist der 29.04.2026.

Definition der Versuchsvariablen

Im Folgenden sind Versuchsvariablen definiert, die für einen besseren Workflow sorgen sollen. Diese werden zum exportieren und überschreiben von Dateien wichtig sein und ermöglichen es, diese Datei für jeden Versuch zu benutzen und lediglich die Variablen zu verändern. Es ist jedoch empfohlen eine Kopie der Forlage für jeden Versuch zu machen, damit dieses Dokument strukturiert bleibt.

Zudem sind wichtige Konstanten definiert, die immer wieder auftauchen.

In [100...

```
versuchsnummer = "256"
versuchsname = "X-ray flourescence"
aufgabe = "0z"
plt_save_folder = f"../img/plots/"
```

Import aller genutzen Libraries

In [101...

```
# Eigene PythonLibrary für das Pap
import Papulator as pap
from Papulator import Colors as c
from Papulator import const
from Papulator import Sympy_Symbols as sym

# Numpy für bessere Berechnungen
import numpy as np
from numpy import exp, sqrt, log, pi
from uncertainties import unumpy as unp

# Weiteres für bessere Rechnungen
import pylab as py

import math

# Berechnungen und Plotting
from scipy import odr
import scipy.optimize
from scipy.optimize import curve_fit
from scipy.stats import norm
from scipy.stats import chi2
from scipy.stats import poisson
from scipy.signal import find_peaks
from scipy.signal import argrelextrema, argrelemin, argrelemax
from scipy.special import factorial
from scipy.integrate import quad

import matplotlib.pyplot as plt
import matplotlib.mlab as mlab
import matplotlib.transforms as transforms

import astropy.units as u

# Zum Auslesen von Dateien und ähnlichem
import os
import os.path

import pandas as pd
import csv
import re

from io import StringIO

# Besseres Funktionen handling
import sympy as sp
from sympy import separatevars
```

Auswertung der Aufgaben

```
In [102... element = ['Fr', 'Mo', 'Ag', 'Cu', 'Ni', 'Ti', 'Zn', 'Zr']
charge_Z = np.array([26, 42, 47, 29, 28, 22, 30, 40])

K_alpha = np.array([6.40, 17.45, 21.80, 8.05, 7.47, 4.67, 8.65, 15.87])
err_K_alpha = np.array([16, 18,19,17,17,16,17,17]) * 1e-2
sqrt_K_alpha=np.sqrt(K_alpha)
err_sqrt_K_alpha =err_K_alpha/(2*np.sqrt(K_alpha))

K_beta = np.array([7.03, 19.54, 24.53, 8.89, 8.26, np.nan, 9.59, 17.96])
err_K_beta = np.array([18, 23,14,20,18,np.nan,16,20]) * 1e-2

sqrt_K_beta=np.sqrt(K_beta)
err_sqrt_K_beta =err_K_beta/(2*np.sqrt(K_beta))
```

```
In [103... n1 = 1
n2 = 2

def fit_func(x,sqrt_Er, sig12):
    return sqrt_Er*(x-sig12)*np.sqrt(1/n1**2-1/n2**2)
```

```
In [104... plt.figure(figsize=(12,7))

plt.errorbar(
    charge_Z,
    sqrt_K_alpha,
    err_sqrt_K_alpha,
    ls=' ',
    capsize=3.5,
    label='Messurment Data'
)

# Regression
popt, pcov=curve_fit(fit_func, charge_Z, sqrt_K_alpha, sigma=err_sqrt_K_alpha)

E_R_rooted = poprt[0]*np.sqrt(1000) # eV
err_E_R_rooted = np.sqrt(pcov[0,0] * 1000)

print(f'E_R rooted= ({pap.round_sig_digs(err_E_R_rooted, E_R_rooted)[2]}) eV')

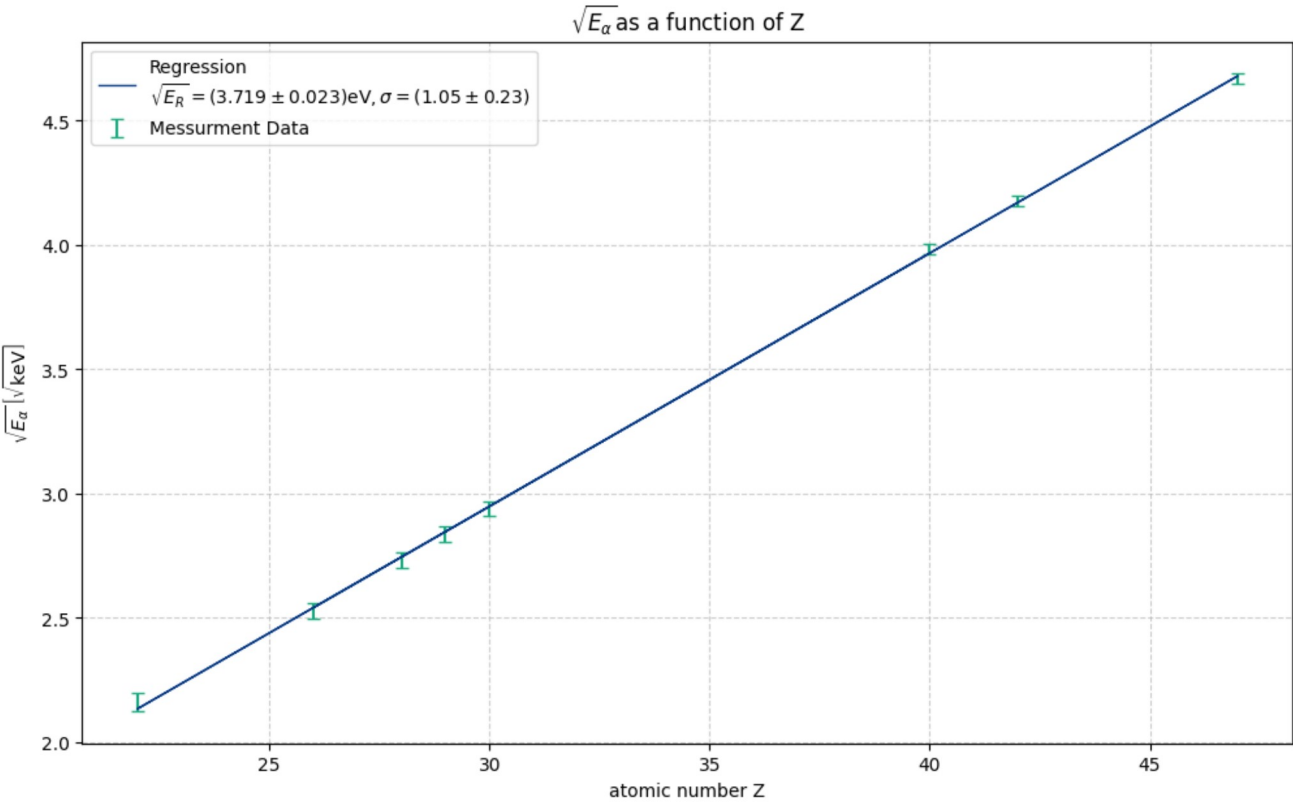
sigma12 = poprt[1]
err_sigma12 = np.sqrt(pcov[1,1])

print(f'sigma = ({pap.round_sig_digs(err_sigma12, sigma12)[2]})')

plt.plot(charge_Z,
    fit_func(charge_Z,*popt),
    label = f"Regression\n$\sqrt{E_R}$ = ({pap.round_sig_digs(err_E_R_rooted, E_R_rooted)[2]}) \mathrm{eV}, \sigma = ({pap.round_sig_digs(err_
    lw=1
)

pap.plot_me(r'$\sqrt{E\_alpha}$'+as a function of Z', 'atomic number Z', r'$\sqrt{E\_alpha} \, , \, \left[\mathrm{\sqrt{keV}}\right]$', f'{plt_save_folder}Al

E_R rooted= (3.719 \pm 0.023) eV
sigma = (1.05 \pm 0.23)
```



```
Out[104... <Axes: title={'center': '$\sqrt{E\_alpha}$as a function of Z'}, xlabel='atomic number Z', ylabel='$\sqrt{E\_alpha} \, , \, \left[\mathrm{\sqrt{keV}}\right]$',
```

```
In [105... E_R = E_R_rooted**2
err_E_R = 2 * err_E_R_rooted * E_R_rooted

print(f'E_R = ({pap.round_sig_digs(err_E_R, E_R)[2]}) eV')
```

```
print(f'standard dviation form lit: {pap.std_abw(E_R, 13.606, err_E_R)} \\sigma')
```

```
E_R = (13.83 \pm 0.17) eV
standart dviation form lit: 1.3 \sigma
```

K Beta

In [106...

```
element = ['Fr', 'Mo', 'Ag', 'Cu', 'Ni', 'Zn', 'Zr']
charge_Z = np.array([26, 42, 47, 29, 28, 30, 40])

K_beta = np.array([7.03, 19.54, 24.53, 8.89, 8.26, 9.59, 17.96])
err_K_beta = np.array([18, 23, 14, 20, 18, 16, 20]) * 1e-2

sqrt_K_beta=np.sqrt(K_beta)
err_sqrt_K_beta =err_K_beta/(2*np.sqrt(K_beta))

# Value for Ti was removed
```

In [107...

```
# For the regression
n1 = 1
n2 = 3

plt.figure(figsize=(12,7))

plt.errorbar(
    charge_Z,
    sqrt_K_beta,
    err_sqrt_K_beta,
    ls= ' ',
    capsize=3.5,
    label='Messurment Data'
)

# Regression
popt, pcov=curve_fit(fit_func, charge_Z, sqrt_K_beta, sigma=err_sqrt_K_beta)

E_R_rooted = pop[0]*np.sqrt(1000) # eV
err_E_R_rooted = np.sqrt(pcov[0,0] * 1000)

print(f'E_R rooted= ({pap.round_sig_digs(err_E_R_rooted, E_R_rooted)[2]}) eV')

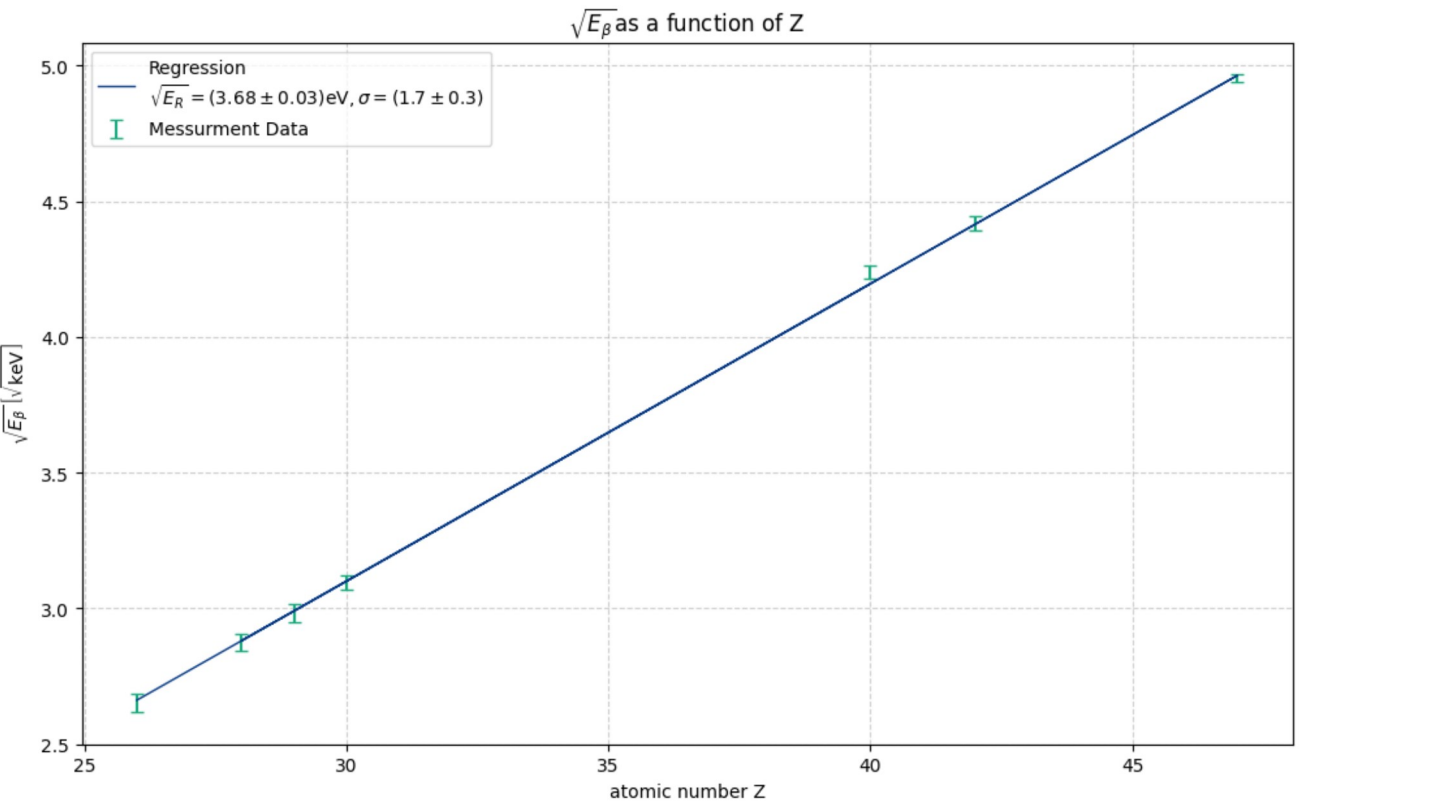
sigma12 = pop[1]
err_sigma12 = np.sqrt(pcov[1,1])

print(f'sigma = ({pap.round_sig_digs(err_sigma12, sigma12)[2]})')

plt.plot(charge_Z,
    fit_func(charge_Z,*popt),
    label = f"Regression\n$\sqrt{{E_R}} = ({pap.round_sig_digs(err_E_R_rooted, E_R_rooted)[2]}) \\\mathrm{{eV}}$, \\\sigma = ({pap.round_sig_digs(err_
    lw=1
)

pap.plot_me(r'$\sqrt{{E_\beta}}$'+as a function of Z', 'atomic number Z', r'$\sqrt{{E_\beta}} \\\, \\\left[\\\mathrm{{\sqrt{{keV}}}}\\\right]$', f'{plt_save_folder}Beta_

E_R rooted= (3.68 \pm 0.03) eV
sigma = (1.7 \pm 0.3)
```



Out[107...

```
<Axes: title='{center': '$\sqrt{{E_\beta}}$as a function of Z'}, xlabel='atomic number Z', ylabel='$\sqrt{{E_\beta}} \\\, \\\left[\\\mathrm{{\sqrt{{keV}}}}\\\right]$', f'ght}$'>
```

In [108...

```
E_R = E_R_rooted**2
err_E_R = 2 * err_E_R_rooted * E_R_rooted

print(f'E_R = ({pap.round_sig_digs(err_E_R, E_R)[2]}) eV')

print(f'standard deviation from lit: {pap.std_abw(E_R, 13.606, err_E_R)} \sigma')
```

```
E_R = (13.51 \pm 0.24) eV
standard deviation from lit: 0.39 \sigma
```

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